

## ALBERT LEA, MN, USA

WMO: 726589

Lat: 43.683N

Lon: 93.367W

Elev: 384

StdP: 96.80

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94968

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |      |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|------|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%   |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS   | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l)  | (m)   | (n)                   | (o)  | (p)   |
| 1             | -24.1      | -22.1 | -27.8                         | 0.3 | -23.6 | -25.6 | 0.4 | -21.5 | 12.0                  | -10.0 | 11.0 | -10.2 | 3.4                   | 300  | 0.544 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.0                   | 32.0            | 23.1 | 30.1 | 22.3 | 28.7 | 21.6 | 25.1                | 29.2 | 24.0 | 28.2 | 23.1 | 27.1 | 4.1                  | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.8                            | 19.5 | 27.5 | 22.6 | 18.2 | 26.7 | 22.1 | 17.6 | 26.3 | 78.4          | 29.2 | 74.5 | 27.9 | 70.7 | 26.9 | 29.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 10.6          | 9.0  | 8.1 | DB                         | -27.8 | 34.4               | 3.5 | 1.9                                                | -30.3 | 35.8       | -32.4 | 36.9       | -34.3 | 37.9       | -36.9 | 39.3 | (4) |
| (5)               |      |     | WB                         | -28.0 | 26.8               | 3.4 | 1.4                                                | -30.5 | 27.7       | -32.4 | 28.5       | -34.3 | 29.3       | -36.8 | 30.3 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 7.8         | -8.9  | -6.8  | 0.2   | 7.9   | 14.6  | 20.3  | 22.4  | 20.8  | 17.2  | 9.6   | 1.6   | -5.7  | (6)  |  |
|                                                                        | DBStd       | 12.32       | 7.33  | 7.07  | 6.95  | 5.76  | 4.97  | 3.71  | 3.04  | 3.02  | 4.66  | 5.35  | 6.32  | 6.60  | (7)  |  |
|                                                                        | HDD10.0     | 2312        | 586   | 471   | 313   | 104   | 14    | 0     | 0     | 0     | 4     | 73    | 259   | 487   | (8)  |  |
|                                                                        | HDD18.3     | 4199        | 844   | 704   | 564   | 315   | 138   | 22    | 4     | 11    | 77    | 275   | 501   | 746   | (9)  |  |
|                                                                        | CDD10.0     | 1523        | 0     | 0     | 8     | 42    | 156   | 309   | 385   | 335   | 219   | 61    | 8     | 0     | (10) |  |
|                                                                        | CDD18.3     | 368         | 0     | 0     | 0     | 2     | 22    | 81    | 131   | 87    | 41    | 4     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 3082        | 0     | 0     | 1     | 34    | 216   | 692   | 1120  | 631   | 341   | 46    | 1     | 0     | (12) |  |
|                                                                        | CDH26.7     | 868         | 0     | 0     | 0     | 9     | 67    | 210   | 341   | 146   | 86    | 9     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 3.6         | 3.9   | 4.0   | 4.1   | 4.4   | 3.9   | 3.2   | 2.7   | 2.4   | 3.0   | 3.6   | 3.9   | 3.8   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 803         | 19    | 18    | 46    | 80    | 103   | 114   | 112   | 101   | 84    | 59    | 40    | 24    | (15) |  |
|                                                                        | PrecMax     | 1287        | 58    | 72    | 105   | 197   | 208   | 256   | 264   | 267   | 231   | 174   | 139   | 69    | (16) |  |
|                                                                        | PrecMin     | 433         | 0     | 0     | 5     | 14    | 31    | 36    | 25    | 17    | 13    | 5     | 1     | 1     | (17) |  |
|                                                                        | PrecStd     | 174         | 14    | 15    | 29    | 42    | 48    | 54    | 62    | 54    | 57    | 38    | 32    | 15    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 7.6   | 12.1  | 22.1  | 27.6  | 32.1  | 33.6  | 33.8  | 32.1  | 32.0  | 27.7  | 20.9  | 12.2  | (19) |  |
|                                                                        |             | MCWB        | 4.3   | 8.7   | 14.8  | 16.3  | 19.9  | 22.7  | 24.8  | 24.3  | 21.9  | 18.5  | 13.4  | 10.0  | (20) |  |
|                                                                        | 2%          | DB          | 4.0   | 7.4   | 17.1  | 22.8  | 28.0  | 31.2  | 32.1  | 29.8  | 28.9  | 23.8  | 17.0  | 7.3   | (21) |  |
|                                                                        |             | MCWB        | 1.9   | 5.3   | 12.5  | 13.6  | 17.6  | 22.0  | 24.3  | 23.3  | 21.4  | 16.1  | 11.8  | 5.2   | (22) |  |
|                                                                        | 5%          | DB          | 2.4   | 4.1   | 12.9  | 20.0  | 26.0  | 28.9  | 30.1  | 27.9  | 27.3  | 21.2  | 12.9  | 4.1   | (23) |  |
|                                                                        |             | MCWB        | 1.2   | 2.4   | 9.3   | 12.2  | 16.7  | 20.5  | 23.6  | 22.3  | 20.0  | 15.2  | 9.1   | 2.4   | (24) |  |
|                                                                        | 10%         | DB          | 1.1   | 2.5   | 9.1   | 17.4  | 22.8  | 27.4  | 28.0  | 27.1  | 25.1  | 17.8  | 10.9  | 2.6   | (25) |  |
|                                                                        |             | MCWB        | 0.3   | 1.1   | 6.6   | 10.9  | 15.2  | 19.8  | 22.2  | 21.6  | 18.8  | 13.0  | 7.7   | 1.3   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 4.8   | 8.5   | 16.4  | 17.6  | 22.4  | 25.0  | 27.5  | 25.7  | 24.1  | 21.4  | 14.8  | 10.0  | (27) |  |
|                                                                        |             | MCDB        | 7.7   | 11.1  | 20.0  | 23.7  | 27.8  | 29.8  | 30.1  | 29.7  | 28.4  | 25.4  | 18.7  | 12.3  | (28) |  |
|                                                                        | 2%          | WB          | 2.5   | 5.2   | 13.1  | 15.3  | 19.9  | 23.6  | 25.6  | 24.5  | 22.8  | 18.2  | 12.3  | 5.2   | (29) |  |
|                                                                        |             | MCDB        | 4.0   | 7.3   | 16.0  | 20.9  | 25.3  | 28.4  | 29.6  | 28.2  | 26.9  | 21.4  | 15.6  | 7.1   | (30) |  |
|                                                                        | 5%          | WB          | 1.3   | 2.9   | 9.8   | 13.4  | 18.4  | 22.4  | 24.5  | 23.5  | 21.5  | 15.8  | 9.9   | 2.9   | (31) |  |
|                                                                        |             | MCDB        | 2.2   | 4.3   | 13.6  | 18.7  | 23.1  | 27.2  | 28.7  | 27.0  | 25.4  | 19.6  | 13.5  | 4.2   | (32) |  |
|                                                                        | 10%         | WB          | 0.0   | 1.2   | 6.9   | 11.6  | 17.0  | 21.1  | 23.5  | 22.4  | 20.1  | 13.7  | 7.6   | 1.5   | (33) |  |
|                                                                        |             | MCDB        | 0.9   | 2.5   | 9.2   | 16.2  | 21.3  | 25.8  | 27.6  | 25.8  | 23.9  | 17.5  | 9.9   | 2.6   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 8.4   | 8.6   | 8.8   | 10.6  | 10.9  | 10.2  | 10.0  | 10.0  | 11.3  | 10.5  | 8.6   | 7.6   | (35) |  |
|                                                                        |             | MCDBR       | 9.5   | 9.4   | 13.4  | 15.9  | 14.0  | 12.2  | 10.9  | 10.8  | 12.8  | 14.5  | 12.5  | 9.3   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 7.9   | 7.5   | 8.9   | 8.3   | 6.2   | 5.5   | 5.5   | 5.7   | 6.2   | 7.6   | 8.1   | 7.5   | (37) |  |
|                                                                        |             | MCWBR       | 8.7   | 8.8   | 12.6  | 13.4  | 11.3  | 10.3  | 9.7   | 9.6   | 10.9  | 11.4  | 11.6  | 8.2   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.266 | 0.288 | 0.311 | 0.366 | 0.393 | 0.415 | 0.417 | 0.404 | 0.374 | 0.326 | 0.289 | 0.266 | (40) |  |
|                                                                        |             | taud        | 2.429 | 2.364 | 2.427 | 2.318 | 2.302 | 2.281 | 2.270 | 2.300 | 2.355 | 2.485 | 2.509 | 2.453 | (41) |  |
|                                                                        | Edn at Noon |             | 877   | 911   | 932   | 898   | 879   | 857   | 851   | 850   | 850   | 857   | 844   | 844   | (42) |  |
|                                                                        |             | Edh at Noon | 73    | 93    | 100   | 122   | 127   | 131   | 131   | 123   | 107   | 83    | 68    | 65    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.78  | 2.85  | 3.75  | 4.63  | 5.43  | 6.07  | 6.37  | 5.36  | 4.28  | 2.79  | 1.84  | 1.34  | (44) |  |
|                                                                        | RadStd      |             | 0.19  | 0.29  | 0.37  | 0.49  | 0.42  | 0.47  | 0.28  | 0.38  | 0.30  | 0.37  | 0.16  | 0.10  | (45) |  |

## Historical Trends

|                              |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                          | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| (46) Station Only            | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (58 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page